

$$\vec{r} = (u+v)\hat{i} + (u-v)\hat{j} + (2uv+w)\hat{k}$$

$$\vec{e}_u = \frac{\partial \vec{r}}{\partial u} ; \quad h_1 = |\vec{e}_u|$$

$$\vec{e}_v = \frac{\partial \vec{r}}{\partial v} \quad h_2 = |\vec{e}_v|$$

$$\vec{e}_w = \frac{\partial \vec{r}}{\partial w} \quad h_3 = |\vec{e}_w|$$

$$\vec{r} = (u+v)\hat{i} + (u-v)\hat{j} + (2uv+w)\hat{k}$$

$$\vec{e}_u = \frac{\partial \vec{r}}{\partial u}$$

$$h_1 = |\vec{e}_u|$$

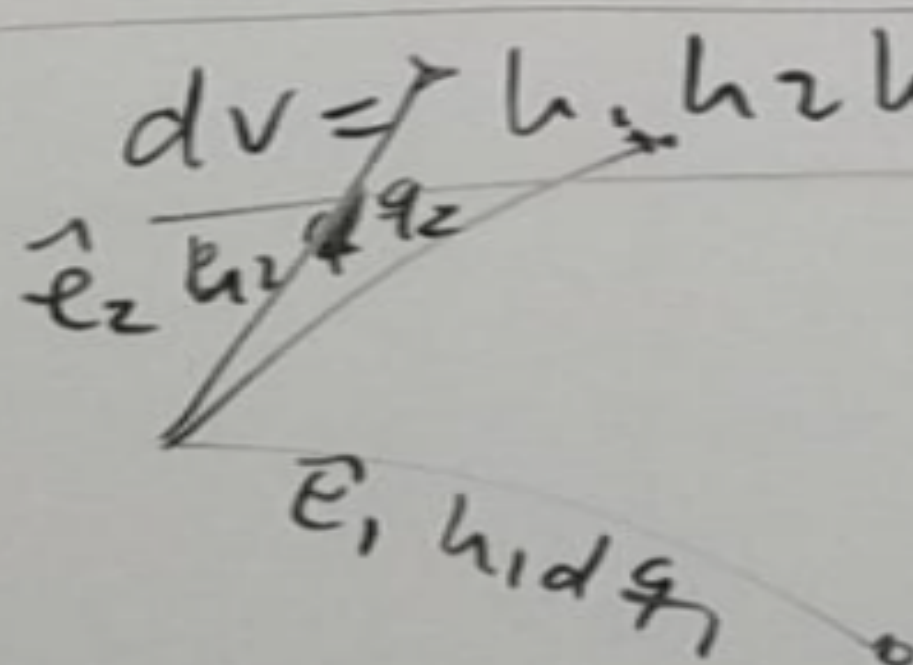
$$\vec{e}_v = \frac{\partial \vec{r}}{\partial v}$$

$$h_2 = |\vec{e}_v|$$

$$\vec{e}_w = \frac{\partial \vec{r}}{\partial w}$$

$$h_3 = |\vec{e}_w|$$

$$\hat{e}_i = \frac{\vec{e}_i}{|h_i|}$$

$$dv = h_1 h_2 h_3 dq_1 dq_2 dq_3$$


$$\hat{e}_1 \times \hat{e}_2 \neq \hat{e}_3.$$

$$h_3 h_1 h_2 d\varphi_1 d\varphi_2 (\hat{e}_1 \times \hat{e}_2) \cdot \hat{e}_3 d\varphi_3.$$

$$ds_1 \rightarrow |\hat{e}_2 \times \hat{e}_3| d\varphi_2 d\varphi_3.$$

$$\vec{L} = i(\vec{r} \times \vec{\nabla}) = i\left(\hat{e}_\sigma \frac{1}{\sin\theta} \frac{\partial}{\partial\theta} - \hat{e}_\theta \frac{\partial}{\partial\sigma}\right).$$

$$\vec{r} \times \vec{\nabla} = \hat{\phi}.$$

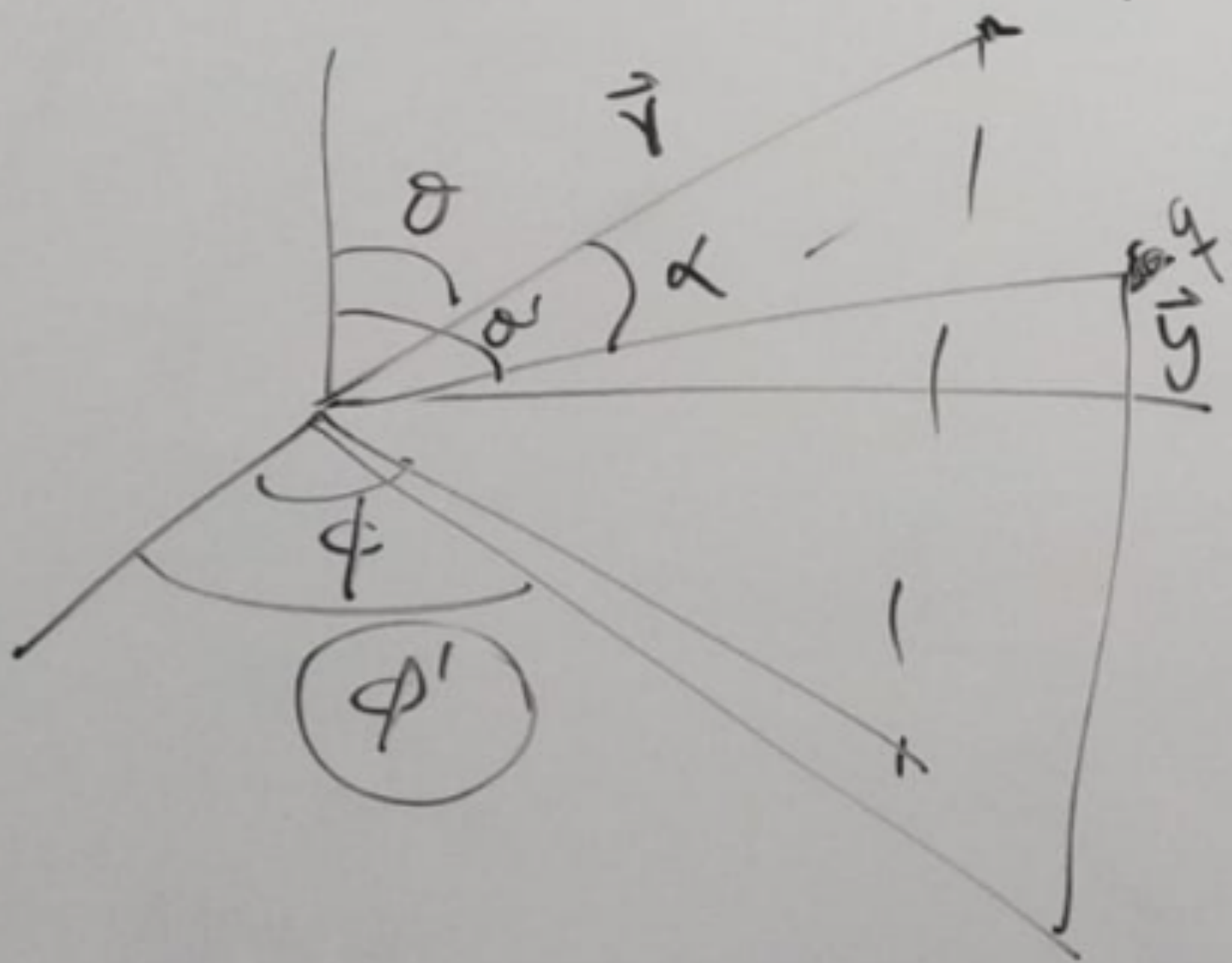
$$L^2 = L_x^2 + L_y^2 + L_z^2 = \frac{1}{\sin^2\theta} \left(\dots \right)$$

$$L_x^2 = L_x \cdot (L_x f)$$

$$\nabla \times \vec{F} = 0$$

$$\phi(r) = \frac{(Q + \frac{q}{y})}{r} - \frac{aq}{y|\vec{r} - \frac{a^2}{y}\hat{y}|}$$

$$\vec{F} = -q\nabla\phi.$$



$$\nabla \frac{1}{|\vec{r} - \vec{r}'|} = \frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|^3}$$

$$\nabla \frac{1}{r} = \frac{\vec{r}}{r^3}.$$

Preparación funciones Bessel

Tenemos el siguiente operador:

$$\mathcal{L}u(x) = p_0 \frac{d^2 u}{dx^2} + p_1 \frac{du}{dx} + p_2 u$$

Bra-Ket

$$\langle u | \mathcal{I} | u \rangle = \int_a^b u \mathcal{I} u \, dx$$

operador adjunto y autoadjunto.

Tenemos

$$\int_a^b u \left(p_0 \frac{d^2 u}{dx^2} + \underline{p_1} \frac{du}{dx} + \underline{p_2} u \right) dx$$

• son funciones

$$u \underline{p_0} \frac{d^2 u}{dx^2} = \frac{d}{dx} \left(u \underline{p_0} \frac{du}{dx} \right) - \frac{d}{dx} (u \underline{p_0}) \frac{du}{dx}$$

Tenemos

$$= \frac{d}{dx} \left(u p_0 \frac{du}{dx} \right) - \frac{d}{dx} \left[\frac{d}{dx} (u p_0) u \right] + \frac{d^2}{dx^2} (u p_0) u$$

Tenemos ...

$$u p_1 \frac{du}{dx} = \frac{d}{dx} (u p_1 u) - \frac{d}{dx} (u p_1) u$$

$$\int_a^b \left\{ \frac{d}{dx} \left(u p_0 \frac{du}{dx} \right) - \frac{d}{dx} \left[\frac{d}{dx} (u p_0) u \right] + \frac{d^2}{dx^2} (u p_0) u + \frac{d}{dx} (u p_1 u) \right\} dx$$

$$- \frac{d}{dx} (u p_1) u + u p_2 u \Big] dx$$

$$\{ \bar{u} | u \}$$

Tenemos

$$u p_0 \frac{d^2 u}{dx^2} + u p_1 \frac{du}{dx} + u p_2 u = \frac{d^2}{dx^2} (u p_0) u - \frac{d}{dx} (u p_1) u$$

$$+ u p_2 u.$$

$$= (u' p_0 + u p_0') u' - (u' p_1 + u p_1') u + u p_2 u$$

$$= (u'' p_0 + u' p_0' + u p_0'' - u' p_1 - u p_1' + u p_2) u$$

$$= (u'' p_0 + 2u' p_0' + u p_0'' - u' p_1 - u p_1' + u p_2) u$$



$$u p_0 u'' + u p_1 u' + u p_2 u.$$

$$u p_1 \frac{du}{dx} = \frac{d}{dx} (u p_1 u) - \frac{d}{dx} (u p_1) u.$$

$$\int_a^b \left\{ \frac{d}{dx} \left(u p_0 \frac{du}{dx} \right) - \frac{d}{dx} \left[\frac{d}{dx} (u p_1) u \right] + \int_a^b \left[\frac{d^2}{dx^2} (u p_1) u - \frac{d}{dx} (u p_1) u + u p_2 u \right] dx + \frac{d}{dx} (u p_1 u) \right\} dx$$

$\langle \bar{I} u | u \rangle.$

$$u p_0 \frac{d^2 u}{dx^2} + u p_1 \frac{du}{dx} + u p_2 u = \frac{d^2}{dx^2} (u p_1) u - \frac{d}{dx} (u p_1) u + u p_2 u$$

$$= (u p_0 + u p_0' u - (u' p_1 + u p_1') u + u p_2 u$$

$$= (u'' p_0 + u' p_0' + u p_0'' + u p_0''') u - u' p_1 u - u p_1' u + u p_2 u$$

$$= (u' p_0 + 2u p_0' + u p_0'') u - u' p_1 u - u p_1' u + u p_2 u$$

$$u p_0 u'' + u p_1 u' + u p_2 u$$

$$0 = - u p_1' u + 2 u p_0' u' + u p_0'' u - 2 u' p_1 u$$

$$2 u (p_0' - p_1) u'$$

$$+ u (-p_1' + p_0'') u = 0$$

$$\underline{p_1 = p_0'}$$

Tenemos (operador adjunto).

$$\bar{I} u = \frac{d^2}{dx^2} (u p_0) - \frac{d}{dx} (u p_1) + p_2 u$$

Tenemos

$$= (u' p_0 + u p_0')' - u' p_1 - u p_1' + p_2 u$$

$$= u'' p_0 + 2u' p_0' + u p_0'' - u' p_1 - u p_1' + p_2 u$$

$$p_0' = p_1$$

$$= u'' p_0 + 2u' p_0' + u p_0'' - u' p_1 -$$

$$\begin{aligned} \mathcal{L} u(x) &= p_0(x) \frac{d^2 u}{dx^2} + p_1(x) \frac{du}{dx} + p_2(x) u \\ \langle u | \mathcal{L} | u \rangle &= \int_a^b u \mathcal{L} u \, dx \\ \mathcal{L} u &= \frac{d^2}{dx^2} (u p_0) - \frac{d}{dx} (u p_1) + p_2 u \\ &= (u' p_0 + u p_0')' - u' p_1 - u p_1' + p_2 u \\ &= u'' p_0 + 2u' p_0' + \cancel{u p_0''} - u' p_1 - \cancel{u p_1'} + p_2 u \\ p_0' &= p_1 \\ &= u'' p_0 + \cancel{u' p_0'} + p_2 u \\ &= \left[\frac{d}{dx} \left(p_0 \frac{du}{dx} \right) + p_2 u \right] \end{aligned}$$

$$a_0 \frac{d^2 u}{dx^2} + a_1 \frac{du}{dx} + a_2 u(x) = 0.$$

$$a_0' \neq a_1.$$

$$e^{\int \frac{a_1}{a_0} dx} \left[\frac{d^2 u}{dx^2} + \frac{a_1}{a_0} \frac{du}{dx} + \frac{a_2}{a_0} u = 0 \right].$$

$$e^{\int \frac{a_1}{a_0} dx} \frac{d^2 u}{dx^2} + e^{\int \frac{a_1}{a_0} dx} \frac{a_1}{a_0} \frac{du}{dx} + e^{\int \frac{a_1}{a_0} dx} \frac{a_2}{a_0} u = 0.$$

$$\frac{d}{dx} \left[e^{\int \frac{a_1}{a_0} dx} \frac{du}{dx} \right] + e^{\int \frac{a_1}{a_0} dx} \frac{a_2}{a_0} u = 0.$$

$$Dy = -\lambda y.$$

$$a_0 \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + (a_2 + \lambda)y = 0.$$

$$\left[\frac{d^2 y}{dx^2} + \frac{a_1}{a_0} \frac{dy}{dx} + \left(\frac{a_2}{a_0} + \frac{\lambda}{a_0} \right) y = 0 \right] e^{\int \frac{a_1}{a_0} dx}.$$

$$\frac{d}{dx} \left[\underbrace{e^{\int \frac{a_1}{a_0} dx}}_{p(x)} \frac{dy}{dx} \right] + \left(\underbrace{\frac{a_2}{a_0} e^{\int \frac{a_1}{a_0} dx}}_{Q(x)} + \underbrace{\frac{\lambda}{a_0} e^{\int \frac{a_1}{a_0} dx}}_{R(x)} \right) = 0.$$

$$p(x) = e^{\int \frac{a_1}{a_0} dx}, \quad Q(x) = \frac{a_2}{a_0} e^{\int \frac{a_1}{a_0} dx}, \quad R(x) = \frac{\lambda}{a_0} e^{\int \frac{a_1}{a_0} dx}.$$

Problema de Eigenvalores

$$\frac{d}{dx} \left[P(x) \frac{dy}{dx} \right] + (Q(x) + \lambda_i R(x)) y_i y_k = 0$$

$$\Rightarrow y_j \frac{d}{dx} \left[P(x) \frac{dy_k}{dx} \right] + (Q(x) + \lambda_k R(x)) y_k y_j = 0$$

$$y_k \frac{d}{dx} \left[P(x) \frac{dy_j}{dx} \right] + (Q(x) + \lambda_j R(x)) y_j y_k = 0$$

$$y_j \frac{d}{dx} \left[P(x) \frac{dy_k}{dx} \right] + (Q(x) + \lambda_k R(x)) y_k y_j = 0.$$

$$y_j \frac{d}{dx} \left[P(x) \frac{dy_k}{dx} \right] - y_k \frac{d}{dx} \left[P(x) \frac{dy_j}{dx} \right] + (\lambda_k - \lambda_j) R(x) y_k y_j = 0.$$

$$(\lambda_j - \lambda_k) \int_a^b R(x) y_k y_j dx = \int_a^b \left\{ y_j \frac{d}{dx} \left[P(x) \frac{dy_k}{dx} \right] - y_k \frac{d}{dx} \left[P(x) \frac{dy_j}{dx} \right] \right\} dx.$$

$$= \int_a^b \frac{d}{dx} \left[P(x) \right]$$

$$y_k \frac{d}{dx} \left[P(x) \frac{dy_j}{dx} \right] + (Q(x) + \lambda_j R(x)) y_j y_k = 0$$

$$y_j \frac{d}{dx} \left[P(x) \frac{dy_k}{dx} \right] + (Q(x) + \lambda_k R(x)) y_k y_j = 0.$$

$$y_j \frac{d}{dx} \left[P(x) \frac{dy_k}{dx} \right] - y_k \frac{d}{dx} \left[P(x) \frac{dy_j}{dx} \right] + (\lambda_k - \lambda_j) R(x) y_k y_j = 0.$$

$$\begin{aligned}
 (\lambda_j - \lambda_k) \int_a^b R(x) y_k y_j dx &= \int_a^b y_j \frac{d}{dx} \left[p(x) \frac{dy_k}{dx} \right] - y_k \frac{d}{dx} \left[p(x) \frac{dy_j}{dx} \right] dx \\
 &= \int_a^b \frac{d}{dx} [p(x)(y_j y'_k - y_k y'_j)] dx \\
 &= p(b)(y_j(b) y'_k(b) - y_k(b) y'_j(b)) \\
 &\quad - p(a)(y_j(a) y'_k(a) - y_k(a) y'_j(a))
 \end{aligned}$$

Tenemos

$$(\lambda_j - \lambda_k) \int_a^b R(x) y_j y_k dx = 0$$

si $\lambda_j \neq \lambda_k$

$$\Rightarrow \int_a^b R(x) y_j y_k dx = 0$$

si $\lambda_j = \lambda_k \Rightarrow \int_a^b R(x) [y_i]^2 dx = 1$

Casos

$$1: \left(y_j(b) y'_k(b) - y_k(b) y'_j(b) \right) = 0$$

$$\left(y_j(a) y'_k(a) - y_k(a) y'_j(a) \right) = 0$$

$$2: p(a) = 0, \quad \left(y_j(b) y'_k(b) - y_k(b) y'_j(b) \right) = 0$$

$$3: p(a) = p(b) = 0$$

$$4: p(a) \neq p(b) \neq 0; \quad y'(a) = y'(b); \quad y(a) = y(b) = 0$$



$$\frac{d^2 y}{dx^2} + \lambda y = 0$$

$$\text{sol. } m^2 + \lambda = 0; \quad e^{mx}$$

$$m = \pm i\sqrt{\lambda}$$

$$y(x) = A \sin \sqrt{\lambda} x + B \cos \sqrt{\lambda} x$$

$$0 = y(0) = A(0) + B$$

$$\Rightarrow B = 0$$

$$y(x) = A \sin \sqrt{\lambda} x$$

$$0 = A \sin \sqrt{\lambda} x$$

$$\sqrt{\lambda} = n\pi$$

$$y(x) = A \sin \sqrt{\lambda} x + B \cos \sqrt{\lambda} x$$

$$0 = y(0) = A(0) + B \Rightarrow B = 0$$

$$y(x) = A \sin \sqrt{\lambda} x$$

$$0 = A \sin \sqrt{\lambda} x$$

$$\sqrt{\lambda} = n\pi$$

$$y_i(x) = A_i \sin i\pi x$$

$$\frac{d}{dx} \left[(1) \frac{dy}{dx} \right] + \lambda y = 0$$

$$p(x) = 1, \quad q(x) = 0, \quad r(x) = 1$$

$$\int_0^1 y_i y_k dx = 0$$

$$\int_0^1 |y_i|^2 dx = 1$$

$$A_k A_j \int_0^1 \sin j \pi x \sin k \pi x dx = 0$$

$$A_k A_j \int_0^1 \sin j \pi x \sin k \pi x dx = 0, \quad j \neq k$$

$$\frac{1}{2} \int_0^1 \left[\cos((j-k)\pi x) - \cos((j+k)\pi x) \right] dx$$

$$\cos j \pi x \cos k \pi x + \sin j \pi x \sin k \pi x$$

$$- \cos j \pi x \cos k \pi x + \sin j \pi x \sin k \pi x$$

$$1 = (y_i)$$

$$2 = p$$

$$3 =$$

$$\frac{1}{2} \left[\frac{\operatorname{sen}(j-k)\pi x}{(j-k)\pi} - \frac{\operatorname{sen}(j+k)\pi x}{(j+k)\pi} \right]_0^1 = 0. \quad 4:$$

Formas autoadjuntas

$$\frac{d}{dx} \left[(1) \frac{dy}{dx} \right] + \lambda y = 0$$

$$|A_j|^2 \int_0^1 \operatorname{sen}^2 \pi j x \, dx = 1$$

$$\frac{|A_j|^2}{2} \int_0^1 (1 - \cos(2\pi j x)) \, dx = 1.$$

$$A_j = \sqrt{2}.$$

$$y_n = \sqrt{2} \operatorname{sen}(n\pi x)$$

$$y_m = \sqrt{2} \cos(n\pi x)$$

$$\mathcal{L} \, u(x) = f(x).$$

$$u(a), u'(a) \\ u(b), u'(b).$$

$$u(x) = \int_a^b G(x,s) f(s) \, ds.$$

$$\mathcal{L} \, G(x,s) = \delta(x-s).$$

$$L = \frac{d}{dx} \left(p(x) \frac{d}{dx} \right) + q(x).$$

$$Lu = \int_a^b L(x,s) f(x) ds.$$

$$\int_a^b \delta(x-s) f(s) ds = f(x)$$